

Topic to discuss

→ Newton divided difference Interpolation.

→ Numerical Problem

→ Homework Problem with
solution.

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Newton divided difference for unequal Interpolation

It is a method use to find an interpolation polynomial when the data points are unequally spaced.

It constructs the polynomial step by step using divided differences, which are based on function values at given points

One major advantage of this method is that new data points can be added easily without recomputing the entire polynomial.

Newton Divided Difference Table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_0	y_0	$\Delta y_0 = \frac{y_1 - y_0}{x_1 - x_0}$	$\Delta^2 y_0 = \frac{\Delta y_1 - \Delta y_0}{x_2 - x_0}$	$\Delta^3 y_0 = \frac{\Delta^2 y_1 - \Delta^2 y_0}{x_3 - x_0}$	$\Delta^4 y_0 = \frac{\Delta^3 y_1 - \Delta^3 y_0}{x_4 - x_0}$
x_1	y_1	$\Delta y_1 = \frac{y_2 - y_1}{x_2 - x_1}$	$\Delta^2 y_1 = \frac{\Delta y_2 - \Delta y_1}{x_3 - x_1}$	$\Delta^3 y_1 = \frac{\Delta^2 y_2 - \Delta^2 y_1}{x_4 - x_1}$	
x_2	y_2	$\Delta y_2 = \frac{y_3 - y_2}{x_3 - x_2}$	$\Delta^2 y_2 = \frac{\Delta y_3 - \Delta y_2}{x_4 - x_2}$		
x_3	y_3	$\Delta y_3 = \frac{y_4 - y_3}{x_4 - x_3}$			
x_4	y_4				

Some Key Concepts

If $(x_0, y_0), (x_1, y_1), (x_2, y_2)$ be given points, the first divided difference for x_0, x_1 is defined by the relation,

$$[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0} = \Delta y_0$$

Similarly, $[x_1, x_2] = \frac{y_2 - y_1}{x_2 - x_1} = \Delta y_1$

Second divided difference of x_0, x_1, x_2 is defined

$$\text{as } [x_0, x_1, x_2] = \frac{[x_1, x_2] - [x_0, x_1]}{x_2 - x_0} = \frac{\Delta y_1 - \Delta y_0}{x_2 - x_0} = \Delta^2 y_0$$

Similarly, third divided difference of x_0, x_1, x_2, x_3

$$\text{is defined as } [x_0, x_1, x_2, x_3] = \frac{[x_1, x_2, x_3] - [x_0, x_1, x_2]}{x_3 - x_0}$$

for n^{th} divided difference,

$$y[x_0, x_1, \dots, x_n] = \frac{y[x_1, x_2, \dots, x_n] - y[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0}$$

Newton divided difference formula

The formula for Newton divided difference is

$$y = f(x) = y_0 + (x-x_0)\Delta y_0 + (x-x_0)(x-x_1)\Delta^2 y_0 + \\ (x-x_0)(x-x_1)(x-x_2)\Delta^3 y_0 + \\ (x-x_0)(x-x_1)(x-x_2)(x-x_3)\Delta^4 y_0 + \dots$$

Numerical Problem: 1

Use Newton divided difference formula to find $f(5)$ from the following data :

x	0	2	3	4	7	8
$f(x)$	4	26	58	112	466	668

Solution : To obtain value of $f(5)$,
we have to make divided difference table,

Also, we have, $x = 5$

x	$y=f(x)$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
0	4	$\frac{26-4}{2-0} = 11$	$\frac{32-11}{3-0} = 7$	$\frac{11-7}{4-0} = 1$	$\frac{1-1}{7-0} = 0$	0
2	26	$\frac{58-26}{3-2} = 32$	$\frac{54-32}{4-2} = 11$	$\frac{16-11}{7-2} = 1$	$\frac{1-1}{8-2} = 0$	
3	58	$\frac{112-58}{4-3} = 54$	$\frac{118-54}{7-3} = 16$	$\frac{21-16}{8-3} = 1$		
4	112	$\frac{466-112}{7-4} = 118$	$\frac{202-118}{8-4} = 21$			
7	466	$\frac{668-466}{8-7} = 202$				
8	668					

Now applying, Newton divided difference formula,

$$y = f(x) = y_0 + (x-x_0)\Delta y_0 + (x-x_0)(x-x_1)\Delta^2 y_0 + \\ (x-x_0)(x-x_1)(x-x_2)\Delta^3 y_0 + \\ (x-x_0)(x-x_1)(x-x_2)(x-x_3)\Delta^4 y_0 + \dots$$

$$= 4 + (5-0) \times 11 + (5-0)(5-2) \times 7 + (5-0)(5-2)(5-3) \times 1$$

$$= 4 + 55 + 105 + 30$$

$$= \underline{\underline{194}}$$

Homework Problem :

find $f(2)$ using Newton divided difference interpolation method, having given

x	4	5	7	10	11	13
$f(x)$	48	100	294	900	1210	2028

Solution : To obtain value of $f(2)$

we have to make divided difference table,

Also, we have, $x = 2$

Newton divided difference table,

x	$y = f(x)$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
4	48	52				
5	100	97	15	1	0	
7	294	202	21	1	0	0
10	900	310	27	1		
11	1210	409	33			
13	2028					

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Now applying divided difference formula,

$$y = f(x) = y_0 + (x-x_0) \Delta y_0 + (x-x_0)(x-x_1) \Delta^2 y_0 + \\ (x-x_0)(x-x_1)(x-x_2) \Delta^3 y_0 + \\ (x-x_0)(x-x_1)(x-x_2)(x-x_3) \Delta^4 y_0 + \dots$$

$$= 48 + (2-4) \times 52 + (2-4)(2-5) \times 15 + \\ (2-4)(2-5)(2-7) \times 1$$

$$= 48 - 104 + 90 - 30$$

$$= 4$$

Ans.

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